Hugo Marquerie 01/11/2025

Lem convergencia uniforme espacio finito imp lp

Lema 1. *Sea* (X, Σ, μ) *un espacio de medida finito y* $(f_n)_{n\in\mathbb{N}}$ *medibles, entonces*

$$f_n \overset{X}{\underset{n\to\infty}{\rightrightarrows}} f \implies \forall p \in [1,\infty) : f_n \xrightarrow[n\to\infty]{\mathcal{L}^p} f.$$

***Demostración*:** Supongamos que $f_n \overset{X}{\underset{n\to\infty}{\rightrightarrows}} f \in \mathcal{L}^\infty(X)$ y $(f_n)_{n\in\mathbb{N}}$ acotada en $\mathcal{L}^\infty(X)$. Sea $M = \max_{n\in\mathbb{N}} \{\|f_n\|_\infty, \|f\|_\infty\} < \infty$. Entonces,

$$|f_n(x) - f(x)|^p \leqslant (|f_n(x)| + |f(x)|)^p \leqslant (2M)^p = 2^p M^p \quad \text{c.t.p. } x \in X.$$

Definimos $g(x) := 2^p M^p$, entonces $g \in \mathcal{L}^1(X)$ porque $\int_X |g| \,\mathrm{d}\mu = 2^p M^p \mu(X) < \infty$. Por el teorema de la convergencia dominada, tenemos que

$$\lim_{n\to\infty} \|f_n - f\|_p^p = \lim_{n\to\infty} \int_X |f_n(x) - f(x)|^p \,\mathrm{d}\mu(x) \overset{\substack{\text{TCD}\\\downarrow}}{=} \int_X \lim_{n\to\infty} |f_n(x) - f(x)|^p \,\mathrm{d}\mu(x) = 0.$$

Por tanto, $\lim_n \|f_n - f\|_p = 0$ y $f_n \xrightarrow[n\to\infty]{\mathcal{L}^p} f$. ■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.